

VASSILY LOMBARD

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Education

Columbia University

BS in Computer Engineering - Richard and Janet Hunter Scholar & Dean's List

Sept. 2025 – May 2027

GPA: 3.79 – New York, NY

Earlham College

BA in Data Science - William Roha Mathematics Award Recipient

Sept. 2022 – May 2025

GPA: 3.97 – Richmond, IN

Relevant Coursework: Artificial Intelligence, Database Management, Data Modeling, Stats & Probability

Experience/Leadership

R&D Pixel Design Intern

STMicroelectronics

May 2026 - Now

Crolles, France

- Design mixed-signal sub-blocks (ADC, sampling stages, and readout chains) for a global shutter pixel array, performing Spectre simulations to optimize circuit specs and ensure full-system functionality.
- Re-architecture the SRAM write-path in Cadence, optimizing transistor sizing and parasitic capacitance to achieve a 2x increase in maximum clock frequency.
- Compile custom hardware testbenches to characterize physical silicon performance in a cleanroom environment, measuring key metrics including dark current and quantum efficiency.
- Achieved the primary goal of the internship by capturing the sensor's first darkroom image, confirming the innovative aspect of the testchip.

Research Assistant

Columbia University UEIL Lab

Nov. 2025 - May 2026

New York City, NY

- Developed quantitative metrics to analyze ultrasound localization microscopy brain images.
- Engineered a bubble tracking algorithm to reconstruct a map of brain vessels accurate at 86%.

Machine Learning Engineer Intern

Palo IT

May 2025 – Aug. 2025

Paris, France

- Optimized the MoE expert selection architecture in Mixtral, improving load balancing across experts by 9%.
- Implemented a local ChatBot with MCP servers to support different skills reducing documentation writing time by 30%.
- Conceived a RAG based app with AWS Lambda and API Gateway to enhance LLM answers for technical documentation.

Robotics Club President

Earlham College

Sept. 2024 – May 2025

Richmond, IN

- Established and led a student-run robotics club with 20+ members.
- Led coding and robot building workshop using VEX IQ and Arduino platforms.
- Organized and participated in the club's first competition where 5 teams built and programmed robots to follow a line.

Projects

2.4-GHz CMOS LNA for Receiver | Cadence Virtuoso, SpectreRF, 130-nm CMOS PDK

Aug. 2026 – Now

- Designed a 2.4-GHz inductively degenerated common-source LNA for a 50 ohms wireless receiver front-end, optimizing transistor sizing, bias current, source degeneration, and LC input/output matching networks.
- Optimized gain, noise figure, impedance matching, and power consumption, targeting > 12 dB gain, < 3.5 dB noise figure and < 10 mW power consumption.
- Developing RFIC layout with on-chip inductors and matching networks, with DRC/LVS, parasitic extraction, and post-layout S-parameter/noise analysis in progress.

BiCMOS Bandgap Voltage Reference | Cadence Virtuoso, Spectre, 130-nm CMOS PDK

Feb. 2026 – Apr. 2026

- Designed a transistor-level 1.2 V bandgap voltage reference using PTAT/CTAT temperature compensation, bipolar devices, current mirrors, and feedback circuitry in Cadence Virtuoso.
- Characterized analog performance using Cadence Spectre, including temperature sweeps, PSRR, noise, power, PVT corners, and Monte Carlo mismatch analysis.
- Verified the physical IC layout, including DRC/LVS and parasitic extraction, and compared pre and post layout performance.

Skills/Interests

Languages: Python, Java, C/C++, Matlab, SQL

Developer Tools: Cadence Virtuoso, Spectre, AWS Cloud, Microsoft Azure (Certified AI-900), GitHub

Libraires: PyTorch, TensorFlow, OpenCV, FastAPI

Activities and Interests: Competitive golf (Scratch handicap), Strength training (315 lb bench)